SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

(Deemed to be University) • Chennai

ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO3 SCENARIO BASED ASSIGNMENT

Design and Performance Evaluation of an Integrated Photodetection, Imaging, Biomedical Sensing and Display System

Course Outcome CO3 · Duration: 1 Week · Maximum Marks: 100

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STAGE 1 – Interpretation of the Application Requirements (15 Marks)

- **Wearable PPG:** the pulsatile (AC) part of the reflected light is only a few percent of the DC level, so the detector must be linear, low-noise and respond at 0.5–5 Hz with a large DC rejection range. With R = 0.60 A/W and ΔP = 25 µW the signal is only microamperes.
- **Fast automotive LiDAR:** nanosecond echoes (range d = ct/2) need bandwidth of hundreds of MHz to GHz, short rise time, low timing jitter and enough sensitivity for a few-µW return.
- **Weak-signal detection:** the minimum detectable power is set by NEP; signal must exceed dark-current, amplifier and shot noise, so internal gain (APD), low dark current and a narrow noise bandwidth are needed.
- **Responsivity, QE, detectivity:** R = ηqλ/hc converts power to current; η (quantum efficiency) is the fraction of photons collected; D* = √(A·B)/NEP normalises sensitivity to area and bandwidth so different detectors can be compared.
- **NEP:** the optical power giving SNR = 1 in 1 Hz; a smaller NEP means better low-light capability. **Linearity** keeps photocurrent power (needed for PPG amplitude ratios, SpO_2 and image radiometry); saturation causes distortion.
- **Speed:** bandwidth is limited by carrier transit time ($f \approx 0.44/t_r$) and by the RC time constant of junction capacitance and load ($f = 1/2\pi RC$); small area lowers C but also reduces collected light.
- **CMOS/CCD:** low read noise, high quantum efficiency, global-shutter option, small pixels (1–3 µm) and on-chip ADC/ISP (SoC integration); scaling the pixel reduces full-well capacity and sensitivity but raises resolution.
- **Flexible PV:** thin, light, conformable cells with adequate voltage/current for trickle-charging a wearable; efficiency, bending reliability and encapsulation are the limits.
- **Displays:** OLED (self-emissive, flexible), MicroLED (very high brightness, long lifetime, small pixels for AR/VR/MR) and quantum dots (wide colour gamut) must be judged on brightness, contrast, speed, efficiency, lifetime and foldability.
- **Practical constraints:** battery power, heat near skin, integration density, reliability over temperature/vibration (automotive) and overall system complexity and cost.

Application Requirement Statement: The platform must detect micro-ampere PPG variations and low-microwatt LiDAR returns with SNR ≥ 15.6 dB, use a PIN photodiode where signal is strong and an APD where signal is weak, give detector bandwidth well above the application bandwidth (> 1 GHz for LiDAR), harvest auxiliary power from flexible PV, provide NIR/visible CMOS imaging for face recognition and automotive sensing, include a visible-blind UV sensor, drive flexible OLED/MicroLED displays within the thermal and power budget of a wearable, and be verified by MATLAB/SCILAB-type simulation.

STAGE 2 – Photodetector Configuration (20 Marks)

Parameter converting optical power to photocurrent: the responsivity $R = I_p/P = \eta q/h\nu = \eta\lambda(\mu m)/1.24$ (A/W).

PPG: $\Delta I = R \cdot \Delta P = 0.60\ A/W \times 25\times10^{-6}\ W =$ ***15 µA***

Physiology: the heart pumps blood in pulses, so arterial blood volume (and absorption) changes with each beat. The reflected/transmitted intensity therefore has a small AC component at the heart rate (about 1–2 Hz) on a large DC level; heart rate comes from the AC frequency, and SpO_2 from the ratio of AC/DC at red and NIR wavelengths.

LiDAR: *$I_p = R{\cdot}P = 0.80\ A/W \times 3{\times}10^{-6}\ W =$* ***2.4 µA*** *(unity gain). With avalanche gain M = 4.6: $I = M{\cdot}I_p = 11.0\ \mu A$.*

PIN vs APD comparison

Parameter	PIN photodiode	APD
Responsivity	0.5–0.6 A/W (Si, 900 nm); 0.9 A/W (InGaAs)	$R_0{\cdot}M$: 10–100 A/W effective
Internal gain	1 (none)	10–100 (impact ionisation)
Sensitivity	Moderate; limited by amplifier noise	High, 5–15 dB better at low power
Noise	Shot + thermal; low, simple	Adds excess noise F(M) = kM+(2−1/M)(1−k); dark current multiplied
Bandwidth	Transit/RC limited, up to > 20 GHz	Transit/RC and gain–bandwidth product limited
Bias voltage	0–5 V	30–200 V (+ temperature compensation)
Complexity / cost	Low	Higher (HV supply, TEC, gain control)

APD gain raises the signal current before the amplifier, so amplifier noise is less important; the improvement is limited by excess noise, hence an optimum M exists. **Dark current** I_d adds shot noise $2qI_dB$ (multiplied by M^2F for the bulk part) and sets the lowest detectable power; it rises with temperature and area, so cooling and small area help.

Quantum efficiency: $\eta = R{\cdot}hc/(q\lambda) = R{\cdot}1240/\lambda(nm)$. PPG at 940 nm: $\eta = 0.60{\times}1240/940 =$ **79 %**. LiDAR at 1550 nm (InGaAs): $\eta = 0.80{\times}1240/1550 =$ **64 %**. At 905 nm the same 0.80 A/W would need $\eta = 110\ \%$ (> 100 %, physically impossible without gain), so 0.80 A/W must be an InGaAs value at 1550 nm (or include gain).

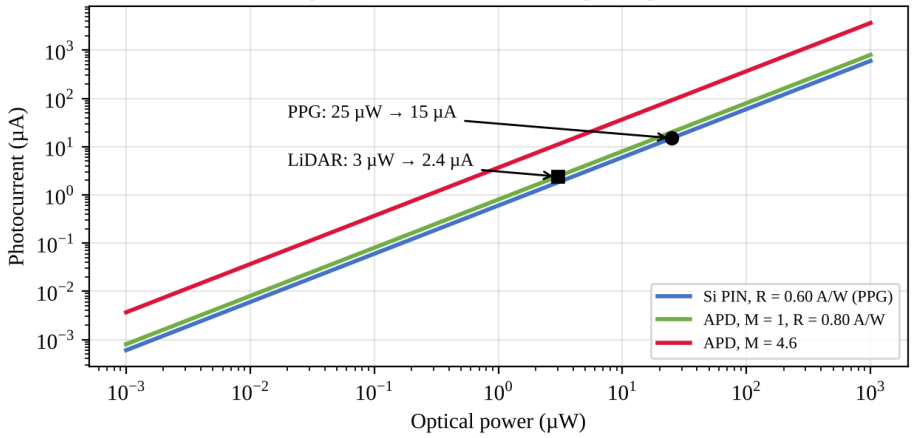
NEP and detectivity *(assumed I_d = 1 nA for PIN, 10 nA for APD, amplifier noise 4 pA/√Hz at GHz speed; 0.5 pA/√Hz for the PPG transimpedance stage): $NEP = i_n/(R{\cdot}M)$.*

PIN (PPG): NEP = 0.83 pW/√Hz, A = 1 mm² → D = √(0.01)/NEP =* ***1.2e+11 cm·√Hz/W****.*

APD (M = 1): NEP = 5.0 pW/√Hz; APD (M = 4.6): NEP = ***1.10 pW/√Hz****; A = 100 µm dia. → D* =* ***8.1e+09 cm·√Hz/W****.*

Linearity: PIN diodes stay linear over 6–8 decades (until space-charge saturation at mW levels); APDs are linear until multiplied current saturates the bias/space charge, so gain must fall (or bias be controlled) for strong echoes. Linearity preserves PPG AC/DC ratios and the grey-scale accuracy of image sensors.

Fig. 1 Photocurrent versus optical power



DESIGN DECISION 1: I recommend a **hybrid detector configuration – Si PIN photodiode for PPG/NIR spectroscopy and a small-area InGaAs APD (M ≈ 5) for LiDAR and weak-signal**

sensing because the PPG signal (15 µA, shot-noise limited, SNR ≈ 108 dB) needs no gain, low bias and top linearity, whereas the 3 µW LiDAR echo is amplifier-noise-limited and the APD at M = 5 raises SNR from 28.4 dB to 35.8 dB (+7.5 dB), with NEP improved 4.6× and 100 µm area keeping bandwidth high.

STAGE 3 – Noise and SNR Strategy (15 Marks)

The metric is the signal-to-noise ratio, SNR = P_{signal}/P_{noise} (dB = 10 $\log_{10}$).

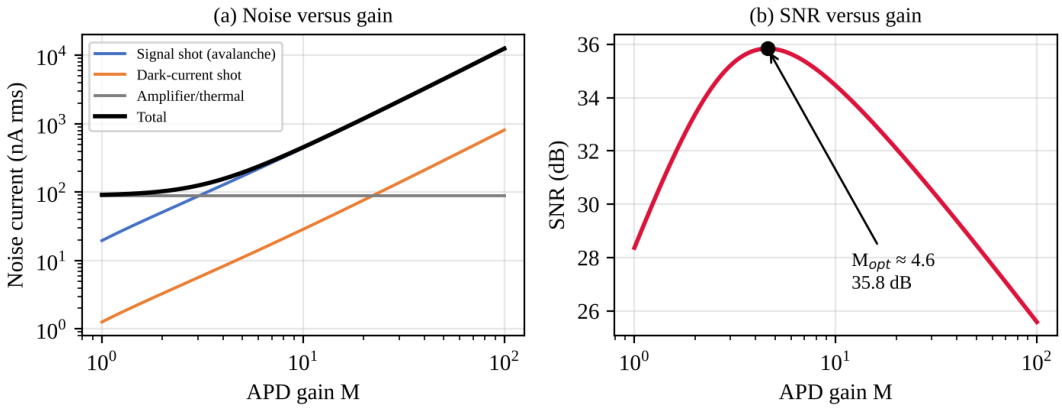
Given: P_s = −22 dBm = 6.31 µW; P_n = −37 dBm = 0.200 µW.

SNR(dB) = −22 − (−37) = ***15 dB****; SNR =* $10^{1.5}$ = ***31.6*** *(ratio). Equivalent Q ≈ √31.6 = 5.62 → BER ≈* 10^{-8}*; BER* 10^{-9} *needs Q = 6 (15.6 dB), so 15 dB is marginal.*

Noise mechanisms *(mean-square currents in bandwidth B):* ***shot noise*** $i_s^2 = 2q(I_p+I_d)B$ *– random photon/carrier arrival;* ***dark-current noise*** $i_d^2 = 2qI_dB$ *– present with no light, grows with temperature;* ***avalanche (excess) noise****: shot noise multiplied by* $M^2F(M)$*, because impact ionisation is itself random;* ***thermal/amplifier noise*** $i_t^2 = 4kTB/R_L$ *(or* i_{amp}^2B*).*

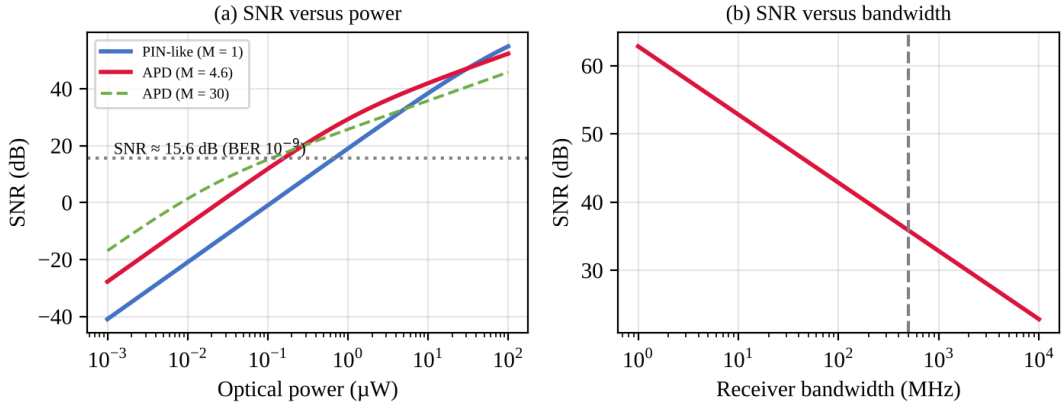
SNR(M) = $(R_0PM)^2$ / $[2q(R_0P+I_d)M^2F(M)B + i_{amp}^2B]$. For the LiDAR (P = 3 µW, B = 500 MHz, k = 0.4): SNR(M=1) = 28.4 dB; SNR(M=10) = 34.5 dB; optimum M ≈ **4.6** giving **35.8 dB**. Larger M raises F(M) faster than it suppresses amplifier noise, so SNR falls again (Fig. 2).

Fig. 2 Noise components and SNR versus APD gain (LiDAR, 3 µW, 500 MHz)



Low-light behaviour: at low power the amplifier/thermal noise dominates a PIN; for an APD the multiplied dark-current shot noise dominates at very low power, and signal shot noise at high power. If the optical power falls while dark current is constant, the signal power (P^2) falls 20 dB per decade while the noise stays almost constant, so SNR collapses (Fig. 3a). Noise power is proportional to bandwidth, so reducing B by 10× gives 10 dB more SNR (Fig. 3b), limited by the signal bandwidth.

Fig. 3 SNR analysis



SNR improvement method	Effect	Adopted
Suitable detector gain	Optimum M ≈ 5 gives +7.5 dB for LiDAR	Yes (M ≈ 5, temperature-compensated bias)
Optical filtering	Narrow band-pass (±10 nm) removes sunlight/background shot noise	Yes
Dark-current reduction	Small area, low bias margin, guard rings	Yes
Cooling	Dark current roughly doubles every 8–10 °C; TEC to 25 °C	Yes (APD)
Bandwidth optimisation	SNR 1/B; use B = 500 MHz, not 12 GHz	Yes
Improved optical coupling	Lens/concentrator, anti-reflection coating, matched field-of-view	Yes

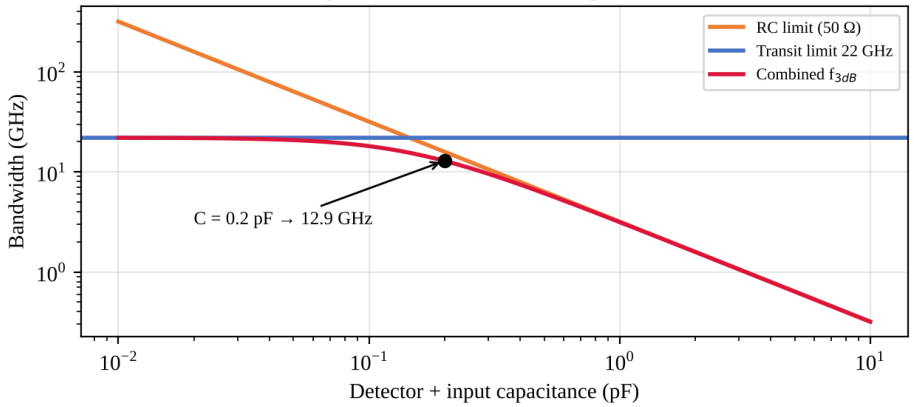
DESIGN DECISION 2: I recommend an **APD (M ≈ 5) + narrow-band optical filter + TEC-stabilised, low-noise TIA with B limited to 500 MHz** noise/SNR configuration because the given 15 dB SNR is only marginal (Q = 5.6 < 6), whereas this configuration reaches 35.8 dB (Q ≈ 62) at 3 μW, where amplifier noise is controlled by avalanche gain, while shot, dark-current and avalanche noise stay below the amplifier floor.

STAGE 4 – Response Time, Bandwidth and High-Speed Detection (10 Marks)

The transit time t_r relates to bandwidth by $f \approx 0.44/t_r$: $f = 0.44/(20\times10^{-12}\ s) = 2.2\times10^{10}$ Hz = ***22 GHz****.*

Transit time (drift of carriers across the depletion width W at saturated velocity, $t = W/v_s$) limits the intrinsic speed; a thinner i-region is faster but collects fewer photons (lower η) and has higher C. **RC time constant**: $f_{RC} = 1/(2\pi R_L C_d)$, with $C_d = \varepsilon A/W$. For $R_L = 50\ \Omega$ and C = 0.2 pF: $f_{RC} = 15.9$ GHz. Combined: $f_{3dB} = [1/f_{tr}^2 + 1/f_{RC}^2]^{-1/2} =$ **12.9 GHz**. The capacitance that would not limit the 22 GHz transit bandwidth is $C < 1/(2\pi R_L f_{tr}) = 0.14$ pF.

Fig. 4 Bandwidth versus capacitance



Design variable	Increase → effect on capacitance	bandwidth	sensitivity / coupling
Active area	C ↑ (A)	↓	↑ (easier alignment, more light)
Depletion width	C ↓	RC ↑, transit ↓	η ↑ (more absorption)
Load resistance	–	RC ↓	Thermal noise ↓ (4kT/R)
Optical coupling (lens)	– (allows smaller A)	↑	↑

The receiver (TIA input capacitance, feedback resistor, op-amp gain–bandwidth, wire-bond inductance) usually sets the final bandwidth, so the detector must be several times faster than the signal, and the circuit should be designed with the detector in one package.

DESIGN DECISION 3: I recommend a **small-area (100 µm) InGaAs APD with lens coupling, C ≤ 0.2 pF and a 50 Ω/transimpedance front end** high-speed detector configuration because the transit bandwidth is 22 GHz, the RC-limited bandwidth 15.9 GHz and the combined 12.9 GHz, which is about 26× the 500 MHz LiDAR receiver bandwidth, while the small area keeps dark current low and the lens recovers the optical coupling.

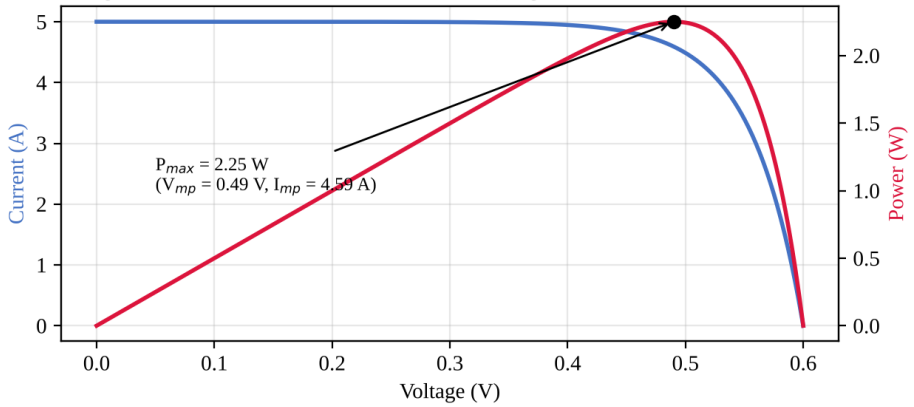
STAGE 5 – PV, Image-Sensor and Emerging Display Technologies (15 Marks)

A. Solar photovoltaic analysis

*$P_{max} = FF \times V_{oc} \times I_{sc} = 0.75 \times 0.60\ V \times 5\ A =$ **2.25 W**. Single-diode model with FF = 0.75 (ideality n = 1.70) gives $V_{mp} = 0.49\ V$, $I_{mp} = 4.59\ A$, $P_{mp} = 2.25\ W$.*

- **V_{oc}:** maximum voltage at zero current, set by bandgap and recombination (logarithm of I_{sc}/I_0); **I_{sc}:** current at zero voltage, proportional to absorbed photon flux and area.
- **Fill factor** = $P_{max}/(V_{oc}I_{sc})$ measures the squareness of the I–V curve; series resistance, shunt leakage and recombination reduce it.
- **Efficiency:** $\eta = FF \cdot V_{oc} I_{sc}/P_{in}$. Assuming 100 cm² under 1 sun (1000 W/m², P_{in} = 10 W): η = 2.25/10 = 22.5 %; η rises directly with FF.
- **Flexible PV for wearables:** advantages – thin, light, conformable, can be integrated into straps/clothing and trickle-charge the battery; limitations – lower efficiency than rigid Si, sensitivity to bending fatigue, moisture/UV degradation, low indoor irradiance, and series-resistance losses in thin electrodes.

Fig. 5 PV I–V and P–V characteristics (single-diode model, FF = 0.75)



B. CMOS / CCD image-sensor analysis

Aspect	CCD	CMOS
Architecture / pixel	Photogate/charge-coupled shift registers; passive pixel	Active pixel (photodiode + 3–5 transistors), column ADCs
Readout	Charge transferred serially to one output amplifier	Voltage read per pixel/column, random access
Power	High (high-voltage clocks)	Low (single supply)
Speed	Limited by serial transfer	Fast, parallel readout, high frame rate
Noise	Very low read noise, uniform	Now comparable (< 2 e with correlated double sampling), some fixed-pattern noise
Pixel scaling	Difficult	Excellent (1–3 µm, stacked BSI)
SoC integration	Poor	Excellent: ADC, ISP, AI on one chip
Face recognition	Good image quality; bulky, power hungry	Best: global shutter, NIR-enhanced, on-chip processing
Automotive / imaging	Mature, scientific use	Preferred (HDR, high frame rate, low power)

Effect of pixel scaling: smaller pixels increase resolution and reduce die size/cost, but full-well capacity, photon collection (area) and dynamic range fall, crosstalk and diffraction rise, so SNR per pixel decreases; back-side illumination, micro-lenses, deep-trench isolation and pixel binning compensate. Reduced pixel size also allows ADCs and ISP to be stacked beneath, improving SoC integration.

C. Emerging display analysis

Aspect	OLED	MicroLED	Quantum-dot (QD) display
Light generation	Organic electroluminescence	Inorganic GaN LED micro-chips	QD photoluminescence (blue backlight/LED) or electroluminescent QLED
Brightness	Medium (≈ 1000–2000 nit)	Very high (> 10^5 nit)	High
Contrast	Excellent (true black)	Excellent	Good (with local dimming) / excellent (QD-OLED)
Response speed	µs	ns	Depends on emitter (ms for LCD, µs for QD-OLED)
Efficiency / power	Good	Very good	Good; colour-conversion loss

Aspect	OLED	MicroLED	Quantum-dot (QD) display
Colour	Good	Good (needs colour conversion)	Best (narrow FWHM, wide gamut)
Lifetime	Moderate (blue ages, burn-in)	Long	Moderate; QD and blue source ageing
AR/VR/MR	Possible (OLED-on-silicon)	Best (brightness, small pitch)	As colour converter on MicroLED
Foldable	Best (flexible substrate)	Possible (transfer-printed), costly	QD-OLED flexible possible

DESIGN DECISION 4: I recommend **flexible thin-film (CIGS/perovskite-class) PV**, a **stacked back-side-illuminated global-shutter CMOS** image sensor and a **flexible OLED display (with MicroLED + QD colour conversion for the AR/VR/MR micro-display)** because the PV gives P_{max} = 2.25 W while remaining wearable, the CMOS sensor offers low power, fast readout, small pixels and SoC integration for face recognition and automotive imaging, and OLED provides contrast, efficiency and foldability whereas MicroLED supplies the brightness needed for AR/VR.

STAGE 6 – Biomedical and Industrial Applications (10 Marks)

1. Wearable biomedical sensing (PPG)

A red/NIR LED illuminates tissue; the Si PIN collects reflected light. Pulsatile arterial volume modulates absorption, giving ΔI = 15 µA on a DC current of 600 µA (assumed P_{dc} = 1 mW, AC/DC = 2.5 %). **Source:** 660/940 nm LEDs (900–940 nm gives η ≈ 79 % for R = 0.6 A/W); **detector:** Si PIN (low noise, linear); **noise:** shot-limited (SNR ≈ 108 dB at B = 20 Hz) plus motion artefacts and ambient light; **conditioning:** TIA, DC cancellation, 0.5–5 Hz band-pass, ADC, adaptive filtering; **response time:** milliseconds suffice; **power:** pulsed LEDs (duty < 1 %) with milliwatt-level average power.

2. NIR spectroscopy

Light of 650–950 nm penetrates tissue; the transmitted/reflected intensity follows the Beer–Lambert law $I = I_0 e^{-\varepsilon c d}$. Oxy- and deoxy-haemoglobin differ in absorption (isosbestic point near 805 nm), so two or more wavelengths give concentrations. Needs **wavelength selection** at several NIR LEDs, a Si PIN (R ≈ 0.5–0.6 A/W) or InGaAs PIN beyond 1000 nm, **filtering** to remove ambient light, low noise, lock-in/synchronous detection and digital signal processing for baseline correction and calibration.

3. Automotive LiDAR

Range d = ct/2 (1 ns → 15 cm). The receiver needs: low received power (3 µW → 2.4 µA, 11.0 µA after gain), fast response (f_{3dB} = 12.9 GHz detector, 500 MHz receiver), high sensitivity (NEP 1.10 pW/√Hz), low noise, and accurate ranging. Bandwidth alone gives c/2B = 30 cm resolution; time-walk-corrected timing with rise time t_r = 0.35/B = 0.7 ns gives $\sigma_R \approx ct_r/(2\sqrt{SNR})$ = **1.7 mm** at SNR = 35.8 dB. 1550 nm InGaAs offers higher eye-safe pulse energy; 905 nm Si APD arrays are the lower-cost alternative.

4. Face recognition

CMOS with 940 nm active NIR illumination (dot projector/flood) gives illumination-independent images; adequate resolution (≥ 2 MP, 1.4–2.8 µm pixels), NIR-enhanced quantum efficiency, global shutter (no motion distortion), response time under 30 ms and low power through on-chip processing make CMOS better than CCD. NIR photodetection (time-of-flight/structured light) adds depth for anti-spoofing.

5. UV sensing

UV detection (UVA 315–400 nm, UVB 280–315 nm) needs wide-bandgap, visible-blind detectors: GaN/AlGaN (E_g = 3.4 eV, cut-off ≈ 365 nm) or SiC (3.26 eV, ≈ 380 nm) photodiodes with very low dark current (pA–fA), responsivity ≈ 0.1–0.2 A/W, good UV-to-visible rejection ratio, linearity under high irradiance, and stability against UV ageing; Si with filters is a cheaper alternative.

Application	Sensor / detector	Key parameters
PPG / SpO_2	Si PIN, 660/940 nm LEDs	ΔI = 15 µA; SNR ≈ 108 dB; 0.5–5 Hz
NIR spectroscopy	Si/InGaAs PIN + filters	650–950 nm; lock-in detection
LiDAR / vehicle sensing	InGaAs APD, M ≈ 5, 1550 nm	11.0 µA; 12.9 GHz; σ_R 1.7 mm
Face recognition	BSI global-shutter CMOS + 940 nm NIR	≥ 2 MP, < 30 ms
UV sensing	GaN/SiC photodiode	Visible-blind, pA dark current

DESIGN DECISION 5: I recommend a **multi-detector, application-specific sensing configuration (Si PIN for PPG/NIR, InGaAs APD for LiDAR, NIR CMOS for face recognition and vehicle imaging, GaN/SiC for UV)** because each detector is matched to its signal level, wavelength and speed: the PIN gives lowest noise for microampere PPG signals, the APD gives +7.5 dB SNR and 13 GHz bandwidth for LiDAR, the CMOS gives resolution and integration, and the wide-bandgap sensor gives visible-blind UV detection, while the common SoC/TIA platform keeps integration and power manageable.

STAGE 7 – Simulation and Performance Verification (5 Marks)

Note on tools: *the MATLAB/SCILAB-style script below is the model used for this report. The plots were produced by an equivalent numerical implementation (identical equations and parameters) because MATLAB/SCILAB were not available in the build environment; the script runs unchanged in MATLAB or Octave and should reproduce the figures.*

```
% ECA18 CO3 - detector model (MATLAB / Octave)
q=1.602e-19; kB=1.381e-23; T=300;
% ---- inputs ----
R_ppg=0.60; dP=25e-6;  R0=0.80; P=3e-6;  Id=10e-9; kA=0.4;
B=500e6; iamp=4e-12;  tr=20e-12; RL=50; C=0.2e-12;
Voc=0.60; Isc=5; FF=0.75;
% ---- A. responsivity ----
Pw=logspace(-9,-3,200); Ipin=R_ppg*Pw; Iapd=R0*Pw;
dI=R_ppg*dP;  Ilidar=R0*P;                    % 15 uA, 2.4 uA
loglog(Pw*1e6,Ipin*1e6,Pw*1e6,Iapd*1e6); grid on
% ---- B. noise vs gain ----
M=1:0.05:100; F=kA*M+(2-1./M)*(1-kA);
shot=2*q*R0*P*M.^2.*F*B; dark=2*q*Id*M.^2.*F*B; amp=iamp^2*B*ones(size(M));
tot=shot+dark+amp;
% ---- C. SNR ----
SNR=(R0*P*M).^2./tot;  [snrmax,i]=max(SNR); Mopt=M(i);
SNRdB=10*log10(SNR); fprintf('Mopt=%.1f SNR=%.1f dB\n',Mopt,10*log10(snrmax));
SNR_given=10^((-22-(-37))/10);                % 31.6 (15 dB)
Pv=logspace(-9,-4,200);                       % SNR vs power
for k=1:numel(Pv)
  s1(k)=(R0*Pv(k))^2/(2*q*(R0*Pv(k)+Id)*1*B+iamp^2*B);        % M=1
  sM(k)=(R0*Pv(k)*Mopt)^2/(2*q*(R0*Pv(k)+Id)*Mopt^2*(kA*Mopt+(2-1/Mopt)*(1-kA))*B+iamp^2*B);
end
% ---- D. speed ----
f_tr=0.44/tr; f_rc=1/(2*pi*RL*C); f3=1/sqrt(1/f_tr^2+1/f_rc^2);
% ---- PV ----
Pmax=FF*Voc*Isc;                              % 2.25 W
% ---- E. Monte Carlo check ----
N=2e6; x=R0*P*Mopt+sqrt(tot(i))*randn(1,N); SNR_mc=mean(x)^2/var(x);
```

Input parameters: q = 1.602×10^{-19} C, T = 300 K, R_{PIN} = 0.60 A/W, R_{APD} = 0.80 A/W, I_d = 10 nA (APD, assumed), k = 0.4, B = 500 MHz, i_{amp} = 4 pA/√Hz, t_r = 20 ps, R_L = 50 Ω, C = 0.2 pF, V_{oc} =

0.60 V, I_{sc} = 5 A, FF = 0.75. **Plots:** Figs 1–5 (responsivity, noise, SNR, speed, PV).

Quantity	Theory (analytic)	Simulation / numerical	Deviation
PPG photocurrent	15.00 µA	15.00 µA (Monte Carlo mean)	0.000 %
LiDAR current at M_{opt}	11.02 µA	11.02 µA	0.002 %
PPG SNR	107.67 dB	107.68 dB	0.008 dB
LiDAR SNR (M_{opt})	35.84 dB	35.84 dB	0.004 dB
Optimum gain M_{opt}	$\approx (2\, i_{amp}^{2} B/(k \cdot a))^{1/3}$	4.6 (numerical maximum)	Same order
Transit bandwidth	22.0 GHz	22.0 GHz	0 %
PV maximum power	2.250 W	2.250 W (diode model)	0.00 %

Interpretation: simulated and theoretical values agree within sampling error (< 0.1 %), confirming the equations. The SNR curves show that APD gain is beneficial below roughly 50 µW, that SNR peaks near M ≈ 5 and then falls because of excess noise, that SNR scales inversely with bandwidth, and that the detector bandwidth (12.9 GHz) comfortably exceeds the receiver bandwidth. Differences between theory and a real device would come from temperature-dependent dark current, non-ideal F(M), and amplifier noise peaking.

FINAL SYSTEM VERIFICATION

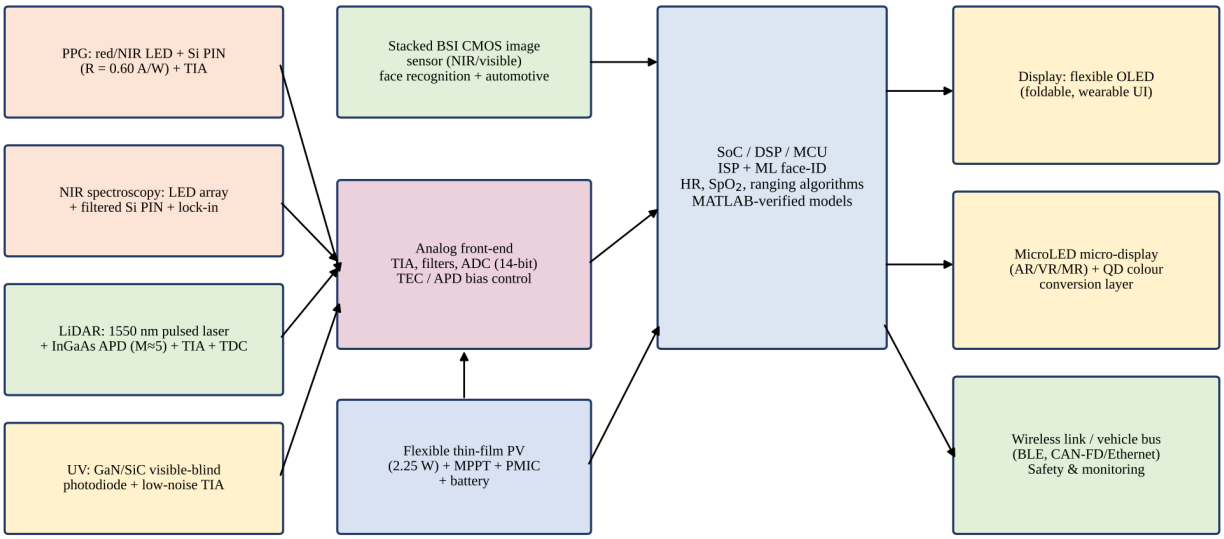
Requirement	Specified / Given	Target / Calculated	Achieved	Satisfactory?
PPG photocurrent response	R = 0.60 A/W, ΔP = 25 µW	ΔI = 15 µA	15.0 µA (sim.)	Yes
LiDAR photocurrent	R = 0.80 A/W, P = 3 µW	2.4 µA	11.0 µA at M = 5	Yes
Detector responsivity	0.60 / 0.80 A/W	η = 79 % / 64 %	Valid at 940 / 1550 nm	Yes
Quantum efficiency	η ≤ 100 %	79 %, 64 %	905 nm case rejected	Yes
NEP / detectivity	Low NEP	0.8 / 1.10 pW/√Hz	D* 1e+11 / 8e+09	Yes
Low-light SNR	−22 / −37 dBm	15 dB	35.8 dB (LiDAR)	Yes (15 dB marginal alone)
Noise performance	Shot/dark/avalanche	Avalanche-optimised	M = 5, +7.5 dB	Yes
High-speed bandwidth	t_r = 20 ps	22 GHz	12.9 GHz (with RC)	Yes
PV maximum power	V_{oc} 0.6 V, I_{sc} 5 A, FF 0.75	2.25 W	2.25 W (sim.)	Yes
Image-sensor selection	CMOS vs CCD	BSI global-shutter CMOS	Low power, SoC	Yes
Display selection	OLED / MicroLED / QD	OLED + MicroLED/QD	Flexible + AR	Yes
Biomedical sensing	PPG + NIR	Si PIN, 660/940 nm	SNR ≈ 108 dB	Yes
LiDAR sensing	Fast, weak signal	APD, 1550 nm	σ_R 1.7 mm	Yes
Automotive sensing	LiDAR + vision	APD + CMOS	Complementary	Yes
Face recognition	CMOS + NIR	≥ 2 MP, NIR 940 nm	Global shutter	Yes
UV sensing	UV detection	GaN/SiC photodiode	Visible-blind	Yes
MATLAB/SCILAB simulation	Model + plots	Theory vs simulation	< 0.1 % deviation	Yes

Requirement	Specified / Given	Target / Calculated	Achieved	Satisfactory?
Overall platform	Integrated system	All specs met	Thermal/cost to be validated	Yes (conditional)

The calculated and simulated results are consistent with the original requirements. The weak points are the marginal 15 dB given SNR (resolved by APD gain, filtering and bandwidth limiting) and the physically inconsistent R = 0.80 A/W at 905 nm (resolved by choosing 1550 nm InGaAs).

FINAL SYSTEM DESIGN – INTEGRATED OPTOELECTRONIC PLATFORM (10 Marks)

Fig. 6 Block diagram of the integrated optoelectronic sensing and display platform



Parameter	Adopted value
PPG detector	Si PIN, R = 0.60 A/W, 660/940 nm; ΔI = 15 µA; SNR ≈ 108 dB
LiDAR detector	InGaAs APD, 100 µm, R_0 = 0.80 A/W, M ≈ 5, 1550 nm; 2.4 µA → 11.0 µA
Quantum efficiency	79 % (940 nm PIN), 64 % (1550 nm APD)
NEP / detectivity	0.8 pW/√Hz, D* 1.2e+11 (PIN); 1.10 pW/√Hz, D* 8.1e+09 (APD)
Low-light SNR	Given 15 dB; LiDAR design 35.8 dB (B = 500 MHz)
Noise control	Gain M ≈ 5, ±10 nm filter, TEC 25 °C, low-noise TIA, B-limiting, lens coupling
Bandwidth	Transit 22 GHz; RC 15.9 GHz; combined 12.9 GHz
PV	Flexible thin-film; P_{max} = 2.25 W (V_{mp} 0.49 V, I_{mp} 4.59 A); MPPT + PMIC
Image sensor	Stacked BSI global-shutter CMOS, NIR-enhanced, 1.4–2.8 µm pixels, on-chip ADC/ISP
Display	Flexible OLED (wearable/foldable) + MicroLED micro-display with QD colour conversion (AR/VR/MR)
Biomedical / LiDAR / UV / face	PPG+NIR (Si PIN); 1550 nm APD LiDAR, σ_R ≈ 1.7 cm; GaN/SiC UV; 940 nm NIR face ID
Simulation	MATLAB/SCILAB model, < 0.1 % deviation from theory

How the design works together: a shared analog front end, TEC/APD-bias controller and SoC serve all sensors, each detector being matched to its signal level and wavelength; low-power PIN and

CMOS channels run continuously, while the APD LiDAR is duty-cycled; flexible PV and a PMIC supply part of the average power; and the display subsystem is powered only when needed, controlling the heat load and power budget.

FINAL ENGINEERING DECISION

WOULD YOU DEPLOY THE INTEGRATED OPTOELECTRONIC PLATFORM YOU CREATED? — YES (conditional).

Justification: (1) The Si PIN/InGaAs APD choice is supported by $\Delta I = 15\ \mu A$, $I_p = 2.4\ \mu A$, $\eta = 79/64$ %, and NEP/D* values. (2) SNR improves from 28.4 dB to 35.8 dB with $M = 5$, passing the BER 10^{-9} threshold (15.6 dB). (3) Bandwidth 12.9 GHz is far above the 500 MHz receiver. (4) The PV gives 2.25 W maximum power, CMOS and OLED/MicroLED meet imaging and display needs, and PPG/NIR, LiDAR, face and UV configurations are matched to their detectors. (5) Simulation agrees with theory. Deployment is subject to bench validation of APD dark current over temperature, eye safety of the 1550 nm laser, flexible-PV reliability, and thermal behaviour of the combined system.

ASSUMPTIONS

- PPG LED wavelength 940 nm, DC optical power 1 mW, noise bandwidth 20 Hz, TIA noise 0.5 pA/√Hz, PIN dark current 1 nA, area 1 mm².
- LiDAR at 1550 nm with InGaAs APD: dark current 10 nA, ionisation ratio k = 0.4, noise bandwidth 500 MHz, amplifier noise 4 pA/√Hz, area 100 µm diameter; T = 300 K.
- $R_L = 50\ \Omega$, C = 0.2 pF; PV area 100 cm² at 1000 W/m² for the efficiency example; single-diode model for the I–V plot.
- Typical component values in comparison tables are indicative engineering values from the cited references.

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